Ethanol Leaf Extract of *Acalypha Wilkesiana* Modulates Streptozotocin-Induced Changes in Electrolytes and Kidney Functions of Diabetic Wistar Rats

[*1]Iyamu A.O., [2]Njoku O.U., [2]Ezeanyika L.A.

[1]Department of Medical Biochemistry, Faculty of Basic Medical Sciences, College of Medicine, Ambrose Alli University, Ekpoma, Nigeria.
[2]Department of Biochemistry, University of Nigeria, Nsukka-Nigeria.

ABSTRACT

Diabetes is a metabolic disorder of insulin deficiency or resistance causing alterations to normal physiology of several organ/system. While we have recently documented ethanol leaf extract of *Acalypha wilkesiana (ELEAW)* to have hypoglycemic effect on the hyperglycemia induced in diabetic rats, there is a dearth of information on the renal physiology, hence the aim of this study. Diabetes was induced using 50mg/kg of freshly prepared streptozotocin (STZ) in 60 adult Wistar rats (170-200g). Thereafter, rats were divided into 3 groups (n = 5) and treated with distilled-water (10ml/kg; group A), glimepiride (0.1mg/kg; group B) and ELEAW (250mg/kg; group C). At baseline, 3 days diabetic induction, 7 and 14 days treatments, body weight and relative weight of the kidneys were determined. Blood was obtained for the estimations of serum creatinine and urea and plasma electrolytes (sodium, chloride, Potassium and bicarbonate). ANOVA was carried out (LSD as the post hoc test) and $p < 0.05$ as significant on the data obtained using SPSS package (version 21). The results revealed increased relative weight of the kidneys, serum creatinine, urea and plasma bicarbonate levels but decreased plasma sodium, potassium and chloride levels in the diabetic untreated group. However, 7 and 14 days treatments with 250mg/kg ELEAW significantly reversed these alterations to near baseline and comparable levels with the standard drug (glimepiride) and control. In conclusion, ELEAW mitigates the diabetic-induced alterations in serum creatinine, urea and plasma electrolytes.

Keywords: *Acalypha wilkesiana*, Diabetes, Renotoxicity, Electrolytes, Kidney function.

***Correspondence. E-mail:** now_splash72@yahoo.com, **Tel.:** +2347035814952

1. INTRODUCTION

Diabetes mellitus is a group of metabolic diseases differentiated by chronic elevation of blood glucose/sugar[1] and disorders in fat and protein metabolism because of deficit insulin secretion/action[2, 3]. The prevalence has risen from 4.7% in 1980 to 8.5% in 2014 in adults[5]. This metabolic disease affects over 246 million people worldwide and was estimated to increase to 380 million by the year 2025 if not given appropriate attention[2, 4]. The disease leads to long term multi-organ complications and causes damage to eye, heart, kidneys, nerves system and blood vessels[2,5-7] with complications in some of these organs leading to death[5, 8-10].

Diabetes is a complicated health condition with multiple treatment options affected by several myths which may influence health-seeking behavior[11]. Currently, diabetes mellitus is managed by a combination of diet, exercise and hypoglycaemic drugs[2,12]. Worrisome, majority of the synthetic drugs/injections have undesirable side effects with many presenting high secondary failure rates[13,14] couple with high purchasing cost especially for people living in developing countries[12,15,16]. These limitations have caused the search for alternative remedies; especially of locally available plant origin. Interestingly, Chan *et al*.[17] had previously documented improved glucose and lipid metabolism and overall condition of diabetic persons with the use of herbal products. In addition, other studies have validated many other plants for diabetic control in experimental animal models[18] and clinical investigations of human patients[19, 20]. We also recently documented hypoglycermic effect of ethanol leaf extract of *Acalypha wilkesiana (ELEAW)* in streptozotocin (STZ)

induced diabetic animal model[21]. Considering therefore the multi-organ alterations implication of diabetic[7], in this study we investigated the effect of ELEAW on the alterations of the kidney induced following diabetes.

2. MATERIALS AND METHODS

2.1. Collection, processing, and extraction of plant materials

Samples of *A. wilkesiana* leaves were collected from Benin City in Edo State, Nigeria, and authenticated at the Herbarium Unit of Forestry Research Institute, Ibadan, Oyo State, Nigeria (FRI.: 110190). The leaves were plucked, sorted, air-dried for 7 days, and then pulverized and packaged in air-tight polyethylene bags. 200g of the powder was added into a container of 1.5L of 70% ethanol and used to prepare the ethanol extract as described by Majekodunmi and Nubani[22].

2.2. Experimental Animals

Sixty (60) adult male Wistar rats (average weight of 170 - 200g) were obtained from the animal house of the College of Medicine, Ambrose Alli University, Ekpoma. The rats were housed at room temperature on a day light and dark night cycle and acclimatized for 14-day with *ad libitum* access to food (rat chow; Vital Feed Nig. Ltd, Jos, Nigeria) and water.

2.3. Induction of diabetes

Diabetes was induced in 24 hrs fasted adult male Wistar rats by intraperitoneal injection with a single dose of 50mg/kg body weight of freshly prepared streptozotocin (STZ) dissolved in 0.10M citrate buffer (pH 4.5). The dose of STZ was based on previous studies by Mythili *et al*.[23] and Szkudelski[24]. To stave off the hypoglycemia during the first day after the STZ injection, diabetic rats were given 5% glucose solution orally as reported by Sugumar *et al.*[25]. Rats with blood glucose levels higher than 300mg/dl after three days (72 hrs) of STZ injection were considered diabetic as previously reported by El-Khateeb[26] and used for the studies.

2.4. Experimental design

The diabetic rats were fasted overnight and divided into three groups (n = 20) and treated as follows:

Group 1: Diabetic control group; treated with 10ml/kg of distilled water.

Group 2: Standard drug group; treated with 0.1mg/kg of Glimepiride.

Group 3: *A. wilkesiana* extract group; treated with 250mg/kg ethanolic leaves extract.

They were treated for 14 days.

2.5. Sample collection

Body weight and blood glucose levels were determined at baseline (before diabetic induction), 3 days after induction of diabetes, 3, 7, 10, and 14 days after treatments. However, 5 rats each were obtained from each group at baseline, 3 days diabetic induction, 7 and 14 days treatment. The rats were cervically decapitated and jugular blood was collected into plain bottle (2 - 3 ml) via accessing the jugular vein. The kidneys were harvested and the weight determined and recorded. All experimental procedures were carried-out following the Guidelines for Care and Use of Laboratory Animals in Biomedical Research[35].

2.6. Sample analysis

The obtained whole body and kidneys weights were used to determined the relative weight of the kidneys and recorded and percentage of body weight.

The serum levels of creatinine and urea were determined using commercial kits (Randox Laboratories kits, United Kingdom) as per the procedure described in the kit. Electrolyte (sodium, potassium, chloride and bicarbonate)

estimations were carried out on plasma using suitable reagents (Fizlab diagnostic) per the procedure described by the manufacturers.

2.7. Statistical Analysis

Data were analyzed using statistical tools from SPSS (version 20) and reported as means ± SEM. One-way analysis of variance (ANOVA) was used to analyze the experimental data and Duncan's multiple test range was used to compare the group means obtained after each treatment. Differences were considered significant when $P \leq 0.05$.

3. RESULTS

Relative weights of the kidneys decrease non-significantly ($p>0.05$) in treated diabetic rats compared to the control (Table 1).

Figures 1 and 2 compare the effects of 14 days treatments with ELEAW, glimepiride and untreated diabetic control on serum creatinine and urea respectively in STZ-induced-diabetic adult male Wistar rats. Diabetes was observed to cause an increase serum creatinine and urea levels. Treatment with 0.1mg/kg glimepiride (1.46±0.23 mg/dL) and 250mg/kg ELEAW (1.27±0.19 mg/dL) resulted in significant decrease in the diabetic-induced increase in serum creatinine. Similarly, 7 and 14 days treatments with 0.1mg/kg glimepiride (1.46±0.23 mg/dL and 1.47±0.24 mg/dL at 7 and 14 days respectively) and 250mg/kg ELEAW (1.27±0.19 mg/dL and 1.07±0.24 mg/dL at 7 and 14 days respectively) resulted in significant decrease ($p<0.05$) in the diabetic-induced increase in serum urea.

Figures 3 to 6 compared the effects of 14 days treatments with ELEAW, glimepiride and untreated diabetic control on plasma sodium (figure 3), potassium (figure 4), bicarbonate (figure 5) and chloride (figure 6) concentrations in STZ-induced-diabetic adult male Wistar rats.

Diabetes resulted in decrease in plasma sodium, potassium and chloride levels while causing an increased in plasma bicarbonate. However, plasma sodium level increased significantly following 7 and 14 days treatments with 0.1mg/kg glimepiride (73.17±7.91 mEq/L and 76.83±9.41 mEq/L at 7 and 14 days respectively) and 250mg/kg ELEAW (71.17±6.24 mEq/L and 74.33±6.40 mEq/L at 7 and 14 days respectively) compared to the untreated control (49.67±5.82 mEq/L and 48.67±5.52 mEq/L at 7 and 14 days respectively). There was a significant increase ($p<0.05$) in plasma potassium concentration with 0.1mg/kg glimepiride (4.97±0.49 mEq/L and 5.38±0.56 mEq/L at 7 and 14 days respectively) and 250mg/kg ELEAW (5.01±0.28 mEq/L and 6.25±0.57 mEq/L at 7 and 14 days respectively) treatments compared to the untreated control (4.46±0.31 mEq/L and 3.82±0.10 mEq/L at 7 and 14 days respectively). Treatment with 0.1mg/kg glimepiride and 250mg/kg ELEAW did not cause any significant change in plasma bicarbonate compared to the untreated control. There was only a significant increase by 14 days treatment with ELEAW compared to the control on plasma chloride.

Table 1. Comparing relative kidney weight in ELEAW, glimepiride and untreated control rats with STZ induced-diabetic.

	Weight of kidneys in grams/Kg body weight			
	Base line	3days post Stz	7days post Rx	14days post Rx
Stz + 10ml/kg Distilled water	2.87	2.86	3.06	3.19
Stz + 0.1mg/kg Glimepiride	2.99	2.92	2.77	2.66
Stz + 250mg/kg A. *wilkesiana*	2.98	2.94	2.82	2.74

Values are in Mean±SEM (standard error of mean), Stz= streptozotocin treated, Rx= treatment, NS indicates not significant ($p>0.05$).

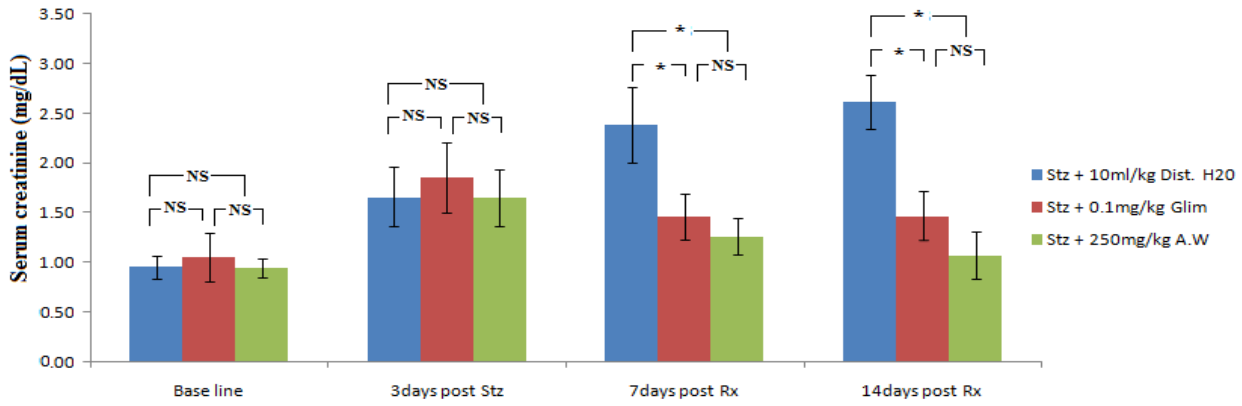


Figure 1. Comparing the effects of 14 days treatment with ELEAW, glimepiride and untreated diabetic control on serum creatinine level of STZ induced-diabetic adult male Wistar rats

Values are in Mean±SEM, *significantly different at p<0.05, NS = not significant (p>0.05), H_2O =water, Dist = distilled, Glim = Glimepiride treated, 250mg/kg = *A. wilkesiana* treated

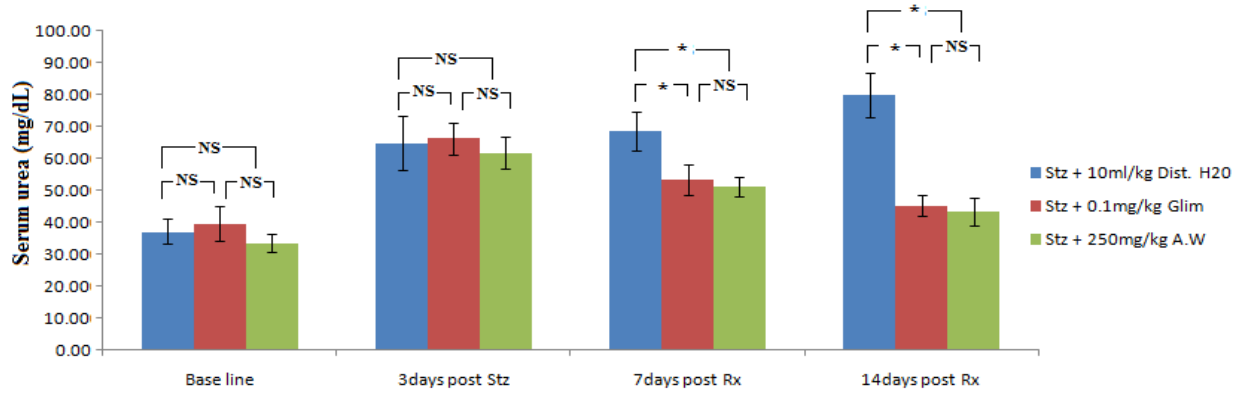


Figure 2. Comparing the effects of 14 days treatment with ELEAW, glimepiride and untreated diabetic control on serum urea levels of STZ induced-diabetic adult male Wistar rats

Values are in Mean±SEM, *significantly different at p<0.05, NS = not significant (p>0.05), H_2O =water, Dist = distilled, Glim = Glimepiride treated, 250mg/kg = *A. wilkesiana* treated

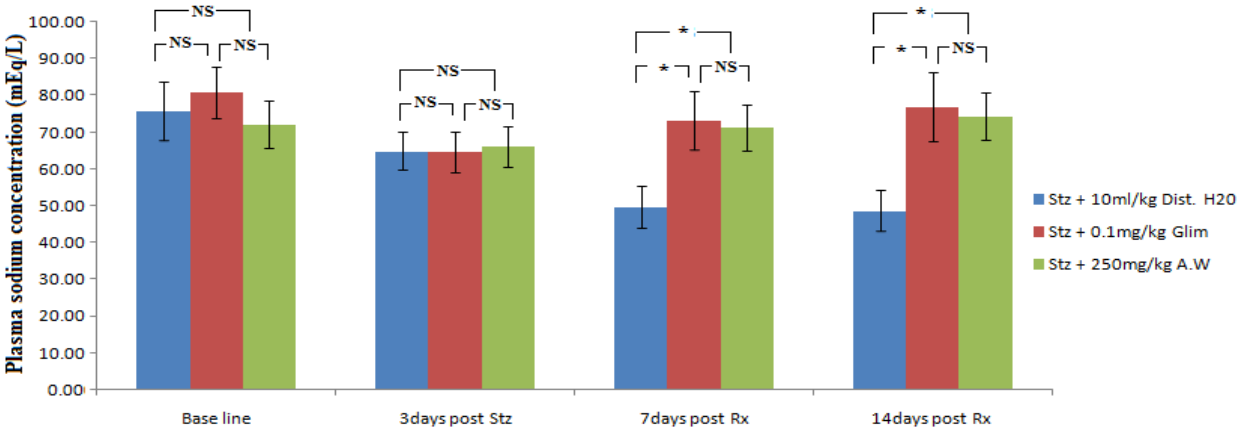


Figure 3. Comparing the effects of 14 days treatment with ELEAW, glimepiride and untreated diabetic control on plasma sodium concentration of STZ-induced-diabetic adult male Wistar rats

Values are in Mean±SEM, *significantly different at p<0.05, NS = not significant (p>0.05), H_2O =water, Dist = distilled, Glim = Glimepiride treated, 250mg/kg = *A. wilkesiana* treated

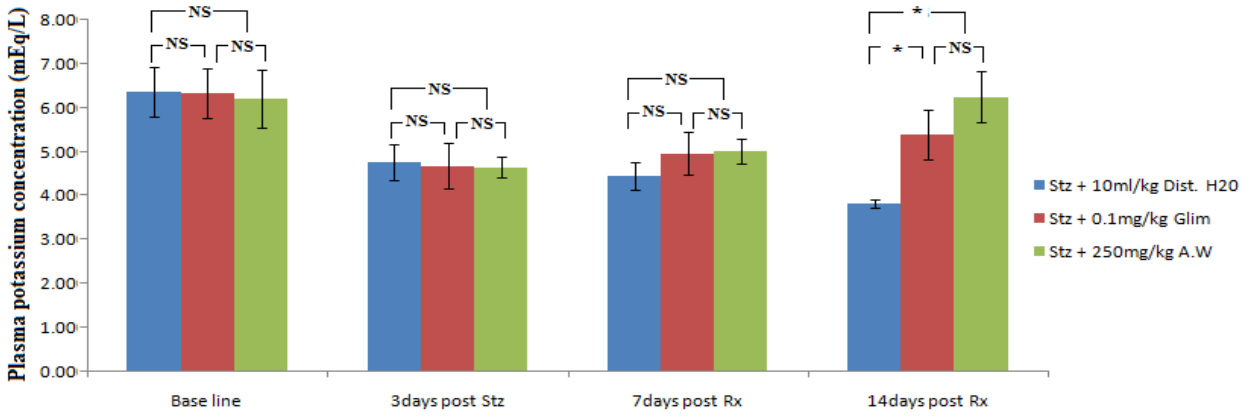


Figure 4. Comparing the effects of 14 days treatment with ELEAW, glimepiride and untreated diabetic control on on plasma potassium concentration of STZ-induced-diabetic adult male Wistar rats

Values are in Mean±SEM, *significantly different at $p<0.05$, NS = not significant ($p>0.05$), H_2O =water, Dist = distilled, Glim = Glimepiride treated, 250mg/kg = *A. wilkesiana* treated

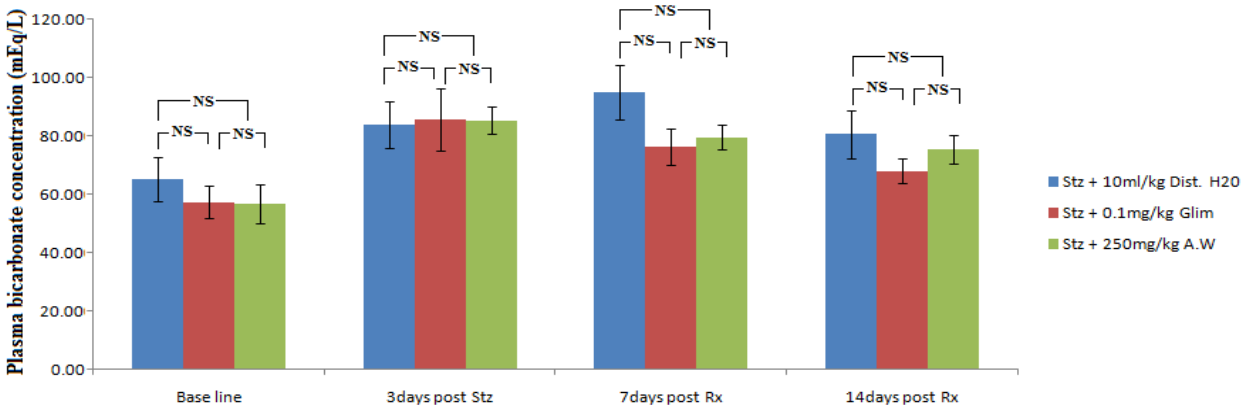


Figure 5. Comparing the effects of 14 days treatment with ELEAW, glimepiride and untreated diabetic control on plasma bicarbonate concentration of STZ induced-diabetic adult male Wistar rats

Values are in Mean±SEM, *significantly different at $p<0.05$, NS = not significant ($p>0.05$), H_2O =water, Dist = distilled, Glim = Glimepiride treated, 250mg/kg = *A. wilkesiana* treated

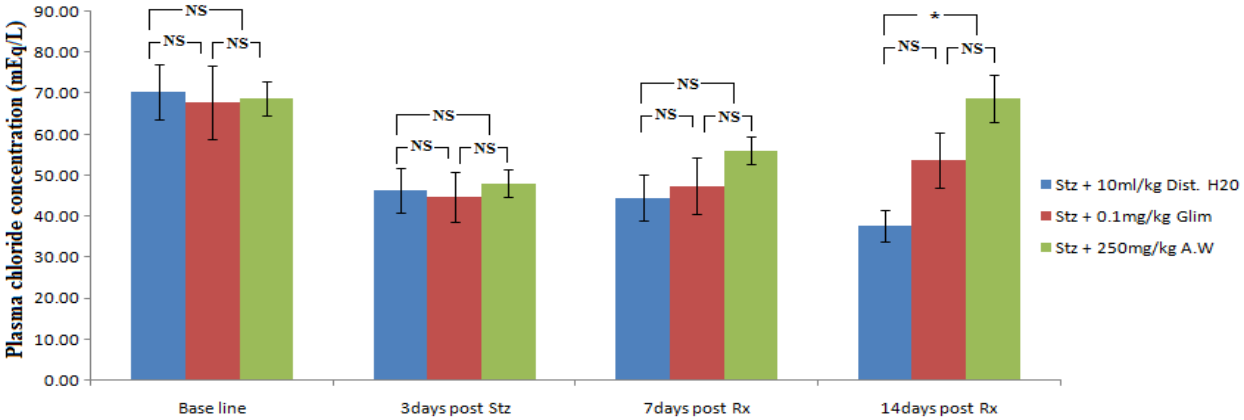


Figure 6. Comparing the effects of 14 days treatment with ELEAW, glimepiride and untreated diabetic control on plasma chloride concentration of STZ induced-diabetic adult male Wistar rats

Values are in Mean±SEM, *significantly different at $p<0.05$, NS = not significant ($p>0.05$), H_2O =water, Dist = distilled, Glim = Glimepiride treated, 250mg/kg = *A. wilkesiana* treated

4. DISCUSSION

There have been increased scientific investigations on the health improvement and benefits of plant parts and extracts. We recently reported ELEAW to mitigate hyperglycemia in diabetic rats[21]. Although diabetes is known to alter the physiology of other organ/system other than blood glucose and metabolism, but there is paucity of literature on the extra-hyperglycemic impact of this plant extract. This necessitated the current investigation to ascertain the anti-nephotoxic properties of the ELEAW on some biochemical parameters of the kidney function in rats. The main function of the kidneys is to excrete the waste products of metabolism and to regulate the body concentration of water and salt[27]. Diabetic hyperglycemia is known to induce elevation of serum levels of urea and creatinine[28] and cause renal dysfunction[29, 30].

As observed in the present study, STZ-induced diabetes caused increase elevation of serum creatinine and urea as well as alters plasma electrolytes. The observed elevations in serum creatinine and urea levels in the present study are in accordance with the findings by El-Khateeb *et al*.[26]. In this regards, Gokce and Haznedaroglu[31], concluded that hormonal and metabolic changes associated with diabetes and the toxic effect of STZ on kidney caused the alteration in kidney function variables. By implication, the alterations in creatinine, urea and plasma electrolytes in the present study may be due to hormonal and metabolic effect by STZ. Alterations in serum electrolytes such as serum Na, K and Ca have also be documented in diabetic rats induced with STZ when compared to control[32]. It was observed in this study that diabetes resulted in decrease plasma levels of sodium, potassium, chloride but elevated plasma bicarbonate concentrations.

However, the observed alterations in serum creatinine and urea levels and plasma electrolytes by in diabetes rats in this study were restored to near baseline levels following 250mg/kg ELEAW treatments in a duration dependent fashion. In line with these observations, El-Khateeb *et al*.[26] and Ikewuchi *et al*.[33] have documented methanolic *A. wilkesiana* leaves extract and aqueous *A. wilkesiana* leaves extract respectively, to correct the diabetes induced elevation in serum creatinine and urea levels. Al-Attar[32] has reported supplementation with aqueous leave extract of *A. wilkesiana* to be beneficial against altered electrolytes profile in experimental diabetic mice. The improvement of the kidney enzymes and electrolytes to near baseline levels associated with 14 days treatment of diabetic rats with ELEAW may be accredited to its phytochemical components and elemental compositions present in the leaf which we also recently evaluated[34]. Thus, the observed anti-diabetic action in our previous study[21] may have resulted in alleviation of altered kidney function and electrolytes status by ELEAW. This may be associated to the plant potent antioxidant content which was also reported in a previous study[36]. Considering that hyperglycaemia increase the generation of free radicals by glucose auto-oxidation and depletes the activity of antioxidant defence system[37], the antioxidant content of ELEAW may have scavenge free radicals and thereby protect renal cells against oxidative stress.

5. Conclusion

Based on the findings of this study, ELEAW mitigates the alterations in kidney function and electrolytes induced following diabetes. Hence, the observed reversal of the altered kidney function and electrolytes markers by ELEAW may have exerted its effect through improving antioxidant defence system and in turn kidney function and electrolytes. This therefore further validates the use of *A. wilkesiana* leave for the treatment of diabetes and its renal complications.

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